

PRACTICAL NO.: 3

Experiment Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/ Tools Required: VS Code (text editor), HTML, CSS, JavaScript.

Theory

JavaScript is a powerful, high-level, interpreted programming language used to create interactive and dynamic web applications. It works together with HTML and CSS, where HTML provides the structure of the webpage, CSS provides the styling, and JavaScript adds functionality such as user interaction, calculations, form validation, and dynamic content updates. JavaScript is widely used for client-side scripting because it executes directly in the user's browser, making web applications faster and more responsive.

One of the most important concepts in JavaScript is **control structures**. Control structures determine the flow of execution of a program by allowing decisions to be made and tasks to be repeated based on specified conditions. They help developers write efficient and logical programs.

The most commonly used control structures in JavaScript are:

- **if Statement:** Executes a block of code only if the specified condition is true.
- **if-else Statement:** Executes one block of code if the condition is true and another block if the condition is false.
- **else-if Ladder:** Used to check multiple conditions one after another until one of them becomes true.
- **switch Statement:** Executes one block of code from multiple available choices based on the value of an expression.
- **for Loop:** Repeats a block of code for a specified number of iterations.
- **while Loop:** Repeats a block of code as long as the given condition remains true.
- **do-while Loop:** Executes the block of code at least once and then continues repeating while the condition is true.

Another important concept in web development is **Form Validation**. Form validation is the process of checking whether the information entered by the user is correct, complete, and within the required limits before processing it. Validation improves data accuracy, enhances user experience, and prevents invalid data from being submitted to the server.

JavaScript performs **client-side validation**, which checks user input before the form is submitted. This reduces unnecessary server requests and provides immediate feedback to the user.

Common validation techniques include:

- Checking that required fields are not left empty.
- Ensuring that only numeric values are entered where required.
- Verifying that marks are within the valid range of **0 to 100**.
- Displaying meaningful error messages when invalid data is entered.
- Preventing form submission until all validation rules are satisfied.

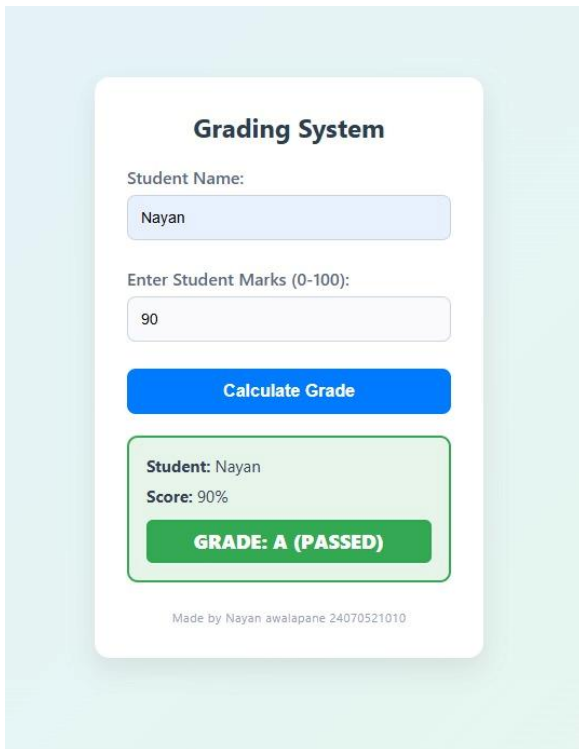
In this experiment, a **Grading System** is developed using HTML, CSS, and JavaScript. The user enters marks through a form, and JavaScript first validates the input to ensure that it is not blank, contains only numeric values, and lies between **0 and 100**. If the entered value is invalid, an appropriate error message is displayed. If the input is valid, JavaScript uses **if-else control structures** to determine the corresponding grade based on predefined grading criteria.

A typical grading system is as follows:

- **90–100:** Grade A+
- **80–89:** Grade A
- **70–79:** Grade B
- **60–69:** Grade C
- **50–59:** Grade D
- **Below 50:** Grade F (Fail)

The calculated grade is then displayed on the webpage without refreshing the browser. This experiment demonstrates the practical use of conditional statements, decision-making, input validation, and dynamic content display using JavaScript. It also helps students understand how real-world web applications validate user input before processing data.

OUTPUT:



The screenshot displays a web application titled "Grading System". It features two input fields: "Student Name:" with the value "Nayan" and "Enter Student Marks (0-100):" with the value "90". A blue "Calculate Grade" button is positioned below the inputs. The results are shown in a green-bordered box: "Student: Nayan", "Score: 90%", and a green button labeled "GRADE: A (PASSED)". At the bottom, a small footer reads "Made by Nayan awalapane 24070521010".

Result

The experiment was performed successfully. The grading system accepted user-entered marks, validated the input correctly, and displayed the appropriate grade based on the specified grading criteria. Invalid inputs such as empty fields, non-numeric values, and marks outside the range of 0–100 were detected successfully, and suitable error messages were displayed.

Conclusion

The experiment was completed successfully. It demonstrated the implementation of **control structures** and **form validation** using JavaScript. The grading system effectively validated user input, applied conditional statements to determine grades, and displayed accurate results dynamically. This experiment enhanced the understanding of decision-making statements, input validation techniques, and interactive web application development using HTML, CSS, and JavaScript.

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